Information Security

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Motivation


فَبِاَیِّ اٰلَآءِ رَبِّکُمَا تُکَذِّبٰنِ
تو (اے جن و انسان!) تم دونوں اپنے رب کی کون کون سی نعمتوں کو جھٹلاؤ گے؟
SURAH REHMAN 55:13

Classical Encryption Techniques

Basic Terminology

- Plaintext
  - The original message

  Ciphertext
  - The coded message
- Enciphering or Encryption
  - Process of converting from plaintext to ciphertext
- Deciphering or Decryption
  - Restoring the plaintext from the ciphertext
- Cryptography
  - Study of encryption
- Cryptanalysis
  (breaking the code)
  - Techniques used for deciphering a message without any knowledge of the enciphering details
- Cryptology
  - Areas of cryptography and cryptanalysis together

Symmetric Encryption

- Also referred to as conventional encryption or single-key encryption
- Was the only type of encryption in use prior to the development of public-key encryption in the 1970s
- Remains by far the most widely used of the two types of encryption.
- 1. symmetric ciphers
- 2. variety of algorithms



Simplified Model of Symmetric Encryption

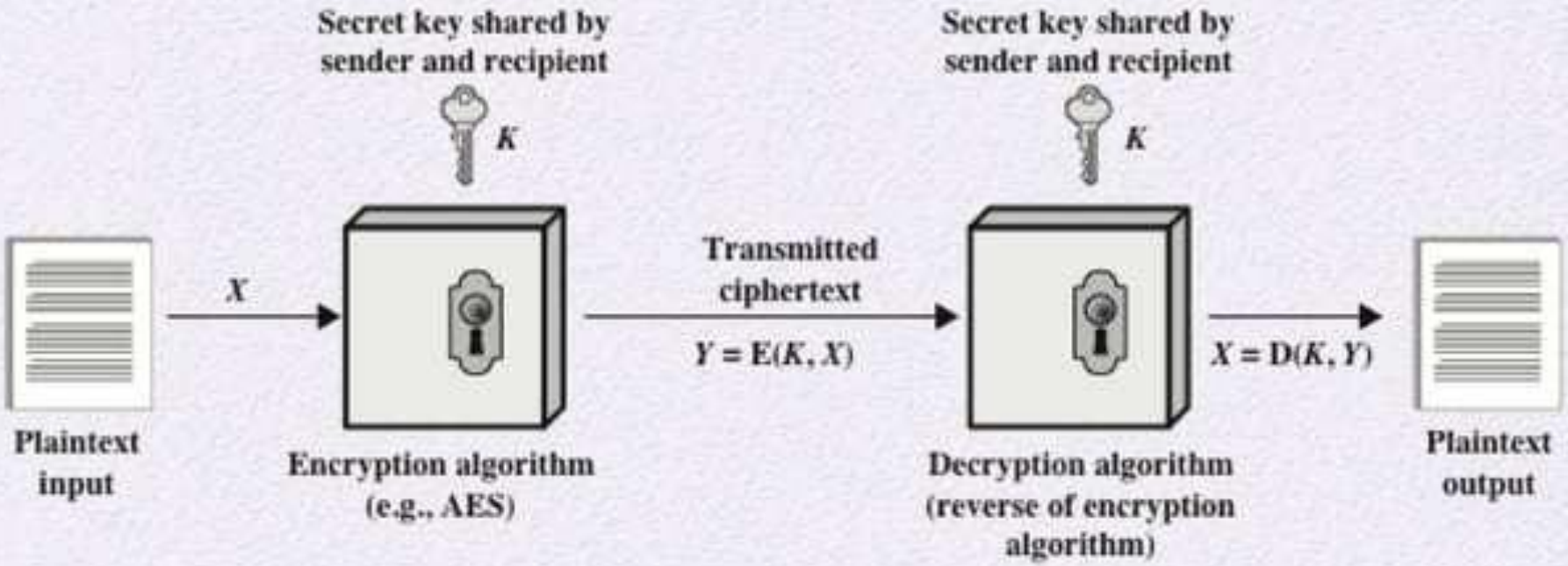


Figure 2.1 Simplified Model of Symmetric Encryption

Symmetric Encryption

A symmetric encryption scheme has five ingredients

- **Plaintext:** This is the original intelligible message or data that is fed into the algorithm as input.

- **Encryption algorithm**: The encryption algorithm performs various substitutions and transformations on the plaintext.

- **Secret key**: The secret key is also input to the encryption algorithm. The key is a value independent of the plaintext and of the algorithm. The algorithm will produce a different output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.

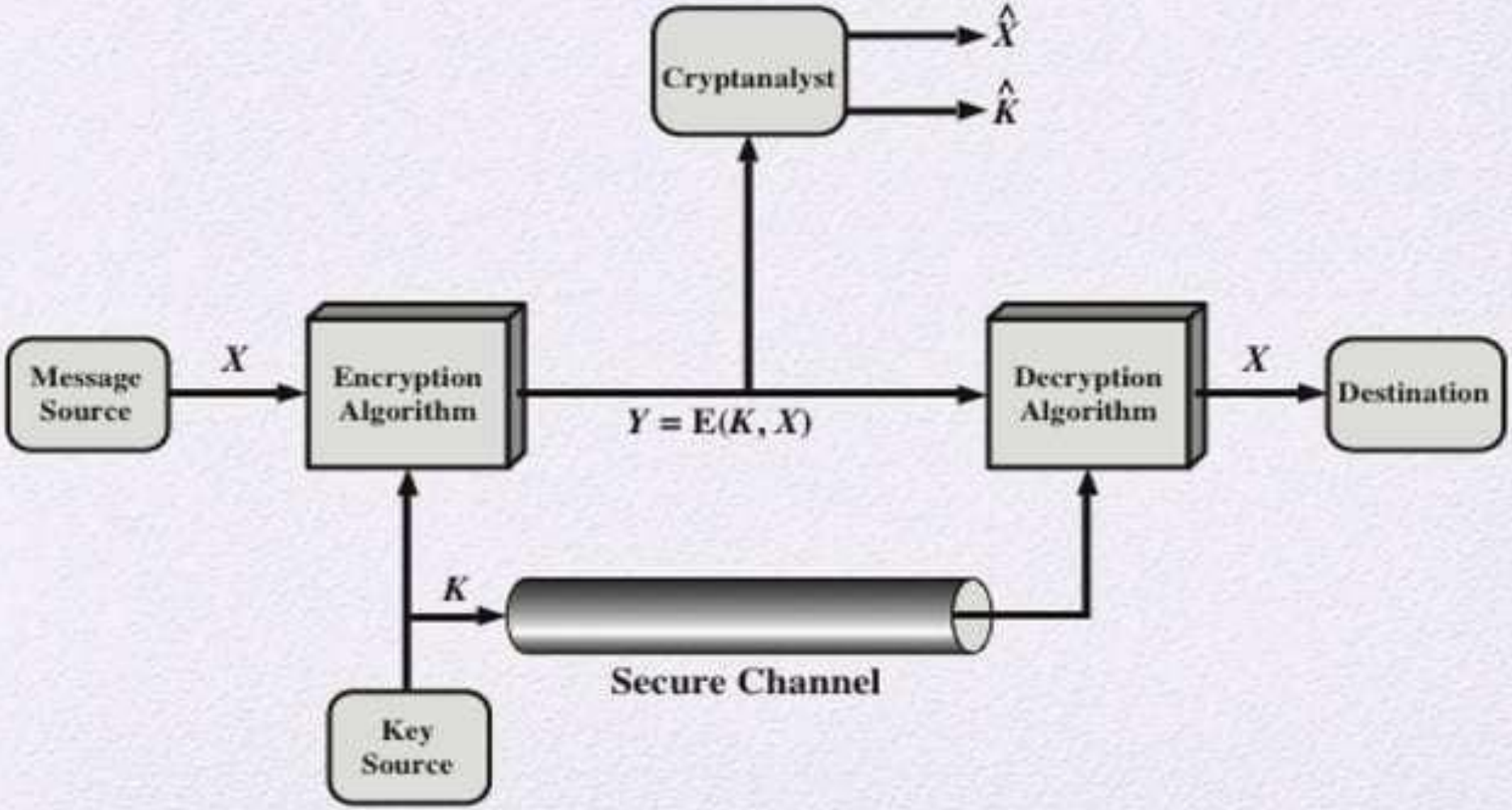
- **Ciphertext:**

This is the **scrambled message** produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts. The ciphertext is an apparently random stream of data and, as it stands, is unintelligible.

- **Decryption algorithm:**

This is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.

Model of Symmetric Cryptosystem



Cryptographic Systems

Characterized along three independent dimensions:

The type of operations used for transforming plaintext to cipher text

- **Substitution**
- **Transposition**

The number of keys used

- Symmetric, single-key, secret-key, conventional encryption (same key)
- Asymmetric, two-key, or public-key encryption (different Keys both sender & receiver)

The way in which the plaintext is processed

- Block cipher -
- Stream cipher

Substitution, in which each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element.

Transposition, in which elements in the plaintext are rearranged

- A block cipher processes the input one block of elements at a time, producing an output block for each input block.
- A stream cipher processes the input elements continuously, producing output one element at a time, as it goes along

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext-ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Table 2.1

Types of Attacks on Encrypted Messages

Encryption Scheme Security

- Unconditionally secure (LONG TIME)
  - No matter how much time an opponent has, it is impossible for him or her to decrypt the ciphertext simply because the required information is not there
- Computationally secure (REQUIRED TIME)
  - The cost of breaking the cipher exceeds the value of the encrypted information
  - The time required to break the cipher exceeds the useful lifetime of the information



Brute-Force Attack

may have a dictionary of all words or commonly used passwords and cycle through those words until it gains access to the account.

A more complex brute-force attack involves trying every key combination until the correct password is found.

Due to the number of possible combinations of letters, numbers, and symbols, a brute force attack can take a long time to complete

1. Substitution Technique

- Is one in which the letters of plaintext are replaced by other letters or by numbers or symbols
- If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns



Playfair Cipher

- Best-known multiple-letter encryption cipher
- Treats digrams in the plaintext as single units and translates these units into ciphertext digrams
- Based on the use of a 5 x 5 matrix of letters constructed using a keyword
- Invented by British scientist Sir Charles Wheatstone in 1854
- Used as the standard field system by the British Army in World War I and the U.S. Army and other Allied forces during World War II

Playfair Cipher (Example)

Perform **Playfair Cipher encryption**.
Plaintext: Pakistan is our Country
Key: MUSLIMS

Step1:- Create the 5x5 Key Matrix. (Fixed always 5 x 5 Matrix)
Remove duplicate letters from the key "MUSLIMS" → MUSLI
Fill in the remaining unused letters of the alphabet (combine I/J as one letter, usually):
So, the new matrix

M U S L I
A B C D E
F G H K N
O P Q R T
V W X Y Z

Playfair Cipher (Example)

Step 2: Prepare the Plaintext
Clean up the plaintext:
Remove spaces and convert to uppercase:
PAKISTANISOURCOUNTRY

Handle repeated letters in digraphs: if a pair has the same letter (like LL), insert an X

Break into digraphs (pairs of 2 letters). If odd number of letters, pad with an extra X

Digraphs:
PA KI ST AN IS OU RC OU NT RY

Playfair Cipher (Example)

Step 3: Encrypt using Playfair Rules
For each digraph:

Same row: Replace each letter with the one to the right (wrap around)

Same column: Replace each letter with the one below (wrap around)

Rectangle: Replace with the letter in the same row but the column of the other letter

Playfair Cipher (Example)

P (Row 4, Col 2), A (Row 2, Col 1) → rectangle → O (4,1) B (2,2) → OB
K (3,4), I (1,5) → rectangle →, N (3,5), L (1,4) → NL
S (1,3), T (4,5) → rectangle → I (1,5), Q (4,3) → IQ
A (2,1), N (3,5) → rectangle → E (2,5), F (3,1) → EF
I (1,5), S (1,3) → same row → M (1,1), L (1,4) → ML
O (4,1), U (1,2) → rectangle → P (4,2), M (1,1) → PM
R (4,4), C (2,3) → rectangle → Q (4,3), D (2,4) → QD
O (4,1), U (1,2) → rectangle → P (4,2), M (1,1) → PM
N (3,5), T (4,5) → same column → T (4,5), Z (5,5) → TZ
R (4,4), Y (5,4) → same column → Y (5,4), L (1,4) → YL
Result:- **OBNLIQEFMLPMQDPMTZYL**

Play Fair Example

To perform **Playfair Cipher encryption.**

Plaintext is **"India is best in IT Field"** and

Key is **"DANGEROUS“**

Reference

[1] cryptography-and-network-security_-principles-and-practice-7th-global-edition by William Stallings